Recommended Assessment

Brushed DC Motor

Driving a Brushed DC Motor

1. What were the differences between rotating the motor by hand when the **`left`** and **`right`** parameters were set to 0.0 in the **dc_motor_raw.py** script and it was executed, versus when the script wasn't running? Why?

2. What directions do the **`left`** and **`right`** parameters drive the brushed DC motor in?

3. What direction does the brushed DC motor turn in when both the **`left`** and **`right`** parameters have non-zero values?

Characterizing a Brushed DC Motor

4. Describe the relationship between the applied voltage and the speed of the motor.

5. What does the **`limitCmd`** parameter do? Why is this important?

Sensing

6. What final **`encoderRatio`** did you use for the provided brushed DC motor? Explain with units.

Modeling

7. Complete the table of motor speeds below based on data collected in your lab.

Voltage (V)	Speed (rad/s)
2	
4	
6	
8	
10	
12	

8. Based on the collected data, estimate the motor speed constant (rad/s/V).

9. (Optional) What does the **`beta`** parameter do and what does this signify?